The Chipathon 2026
Proposal Review: Group 1 A1-A25

IEEE Solid-State Circuits Society
Technical Committee on the Open Source Ecosystem (TC-OSE)

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[A10] CryptoAccel

ASCON AEAD128 based Cryptographic Hardware Accelerator with AXI Interface

as a plug and play hardware co-processor for various applications.

Team members

Discord name	Affiliation	Experience	Role
lakvlsi_90908	IIT, Bombay	Ph.D. Research Scholar	Team Lead / Presenter
zysteresis	MIT, Manipal / STMicroelectronics	Undergraduate (III)	Team Member
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harshi070852	PESIT, Bangalore	Undergraduate (IV)	Team Member

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PROJECT INFORMATION

GOAL:

- Design and implement a lightweight ASCON-AEAD128a cryptographic hardware accelerator with an AXI-Lite interface using GF180MCU technology node
- Enable seamless integration of the accelerator with a wide range of processors and System-on-Chip (SoC) platforms that support the AXI protocol.

DESIGN:

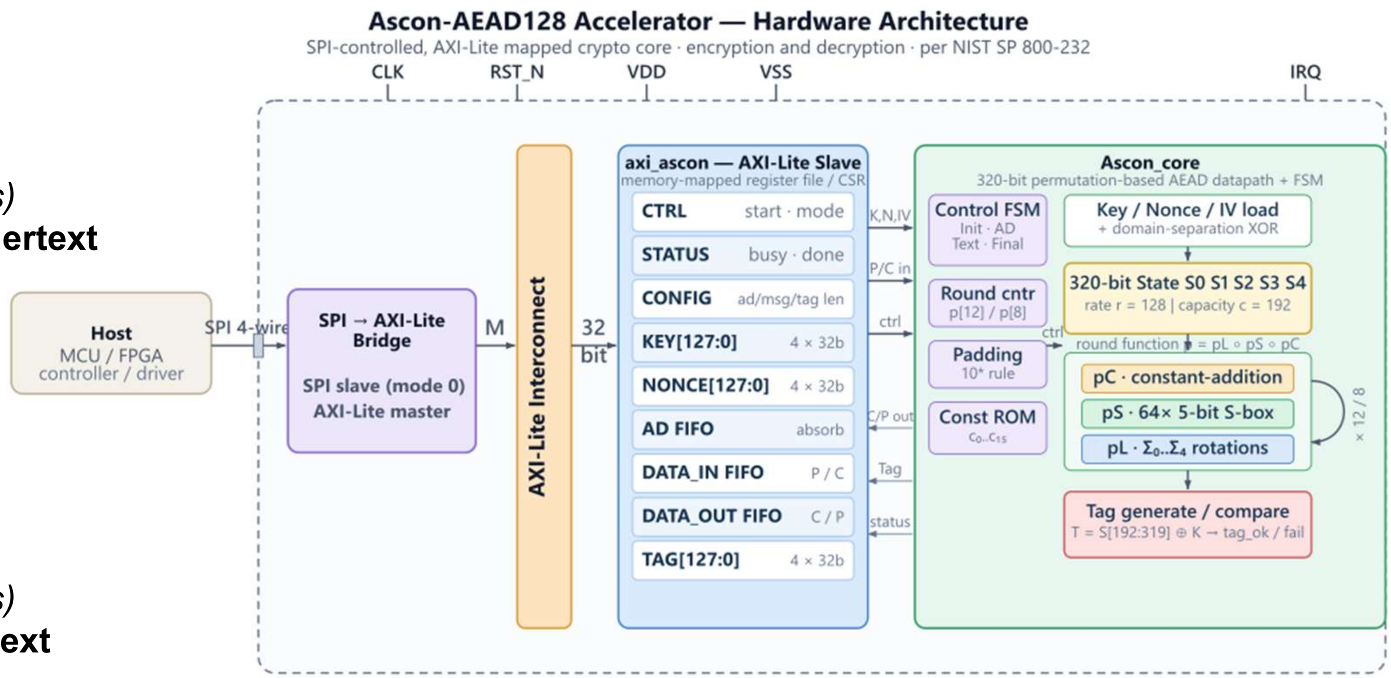
❑ Encryption:

1. **Configure and key the engine** *(SPI writes)*
2. **Start** *(SPI write)*
3. **Feed the associated data** *(SPI writes)*
4. **Stream plaintext and collect ciphertext** *(SPI writes + reads, interleaved)*
5. **Finish and read the tag** *(SPI reads)*

❑ Decryption:

1. **Configure and key the engine** *(SPI writes)*
2. **Start** *(SPI write)*
3. **Feed the associated data** *(SPI writes)*
4. **Stream ciphertext and collect plaintext** *(SPI writes + reads)*
5. **Verify** *(SPI read)*

ARCHITECTURE



APPLICATIONS:

- Internet of Things (IoT) devices: authenticated encryption and data integrity protection for sensor nodes, edge devices, and embedded controllers operating under strict area and power constraints.
- Serve as a hardware Root of Trust for secure boot applications by verifying the authenticity and integrity of firmware during system startup.

SPECIFICATIONS

Technology	GF180MCU
Supply Voltage	3.3V
Frequency	50MHz
AREA	<0.65 μm^2
PIN COUNT	~ 15 PINS

REFERENCES

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[2] K. -D. Nguyen, T. -K. Dang, B. Kieu-Do-Nguyen, C. -K. Pham and T. -T. Hoang, "Live Demonstration: ASIC Implementation of ASCON Lightweight Cryptography for IoT Applications," 2025 IEEE International Symposium on Circuits and Systems (ISCAS), London, United Kingdom, 2025, pp. 1-1, doi: 10.1109/ISCAS56072.2025.11044102.

[3] J. Kaur, A. C. Canto, M. M. Kermani, and R. Azarderakhsh, "A Comprehensive Survey on the Implementations, Attacks, and Countermeasures of the Current NIST Lightweight Cryptography Standard," Electronics, vol. 13, no. 22, Art. no. 4550, 2024.

[4] OpenTitan Project, "EarlGrey and Darjeeling Architectures,“
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